| Question | | | | Marking details | Marks available | | | | Maths | Prac |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | | |
| 9 | *(a)* | (i) | | Use of intensity = $\frac{P}{4\pi R^2}$ or correct substitution (1)<br>Intensity = 1 415 [W m$^{-2}$] (1) | 1 | 1 | | 2 | 2 | |
| | | (ii) | | [Using 1 400 W m$^{-2}$]:<br>20% = 280 W m$^{-2}$<br>15% = 42 W m$^{-2}$ (1) **for both** efficiency calculations<br>Area = $\frac{8 \times 10^9}{42}$ (**ecf**) = $1.9 \times 10^8$ m$^2$ (1) **(unit)**<br>Accept 190 km$^2$ or equivalent | | 2 | | 2 | 2 | |
| | | (iii) | | Greater intensity of radiation in Portugal (accept increased/longer sunshine hours throughout year)<br>Do not accept more efficient | | | 1 | 1 | | |
| | *(b)* | (i) | | $\lambda_{max}T$ = constant | 1 | | | 1 | | |
| | | (ii) | | $\lambda_{max} = \frac{0.0029}{290}$ (1)<br>$\lambda_{max} = 10 \times 10^{-6}$ [m] so in IR (1) | | 2 | | 2 | 2 | |
| | | (iii) | I | <u>Surface emits IR</u> in region 5 μm to 20 μm (1)<br>[Partially] <u>absorbed by water vapour</u> (and $CO_2$) (1)<br>Correct reference to numerical data from graphs e.g. either water: greatest absorption between 5 μm to 8 μm **or** 12 μm to 20 μm) **or** $CO_2$: (greatest absorption between 10 μm to 20 μm) (1)<br>[IR] re-emitted in all directions (1) | 1<br>1 | | 1<br>1 | 4 | | |
| | | | II | Higher temperatures will lead to increased water vapour in atmosphere (1)<br>Leading to further absorption and increase in temperature (1)<br>**Alternative:**<br>As the ice cap melts leaving darker surface (1)<br>which absorbs more visible radiation and increase in temperature (1)<br>**OR**<br>Melting permafrost (1)<br>means more methane in atmosphere (1) | | | 2 | 2 | | |



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| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | *(c)* | (i) | | $U$ value = Energy per second / (area × temperature difference) / rate of heat transfer per unit area per unit temperature difference / $U = \frac{P}{A\Delta\theta}$ with symbols correctly identified (1)<br>Thickness of material **or** conductivity of material (1) | 2 | | | 2 | | |
| | | (ii) | | Energy per second lost = 1.8 × (5 × 3)× (21 – 9) (1) for substitution<br>Energy per second = 324 [W] (1) | | 2 | | 2 | 2 | |
| | | (iii) | | 1 000 = 1.8 × 15 × (21 – $T$) (1)<br>$T$ = -16 [°C] (1) | | 2 | | 2 | 2 | |
| | | | | **Question 9 total** | **6** | **9** | **5** | **20** | **10** | **0** |



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